

A Survey and Reader’s Guide to UGP Physics

Programme Overview, Paper Summaries, and Reading Paths

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Abstract

The **Reflexive Reality** research programme is a sustained formal investigation into the structural constraints imposed on any perfectly self-contained reality — one with no “outside” from which its laws could be externally selected. The programme encompasses physics, chemistry, biology, evolutionary systems, computational systems, cognitive systems, and AI. It comprises two sister sub-programmes: **NEMS** (No External Model Selection), which formalizes the “no outside” constraint and machine-checks its theorem-grade consequences in Lean 4; and the **UGP Physics Programme**, the quantitative physics layer that derives masses, couplings, and quantum numbers from within. The central NEMS results (gauge group forced, three generations selected, Born rule derived, arrow of time fixed, quantum gravity constrained) are archived in 93+ papers at <https://www.novaspivack.com/research>.

The **UGP Physics** corpus is the *quantitative physics sub-programme* of Reflexive Reality, sibling to NEMS. Where the NEMS papers establish *why* the Standard Model structure is the only consistent option, the UGP papers ask *what numerical values* the masses, couplings, and quantum numbers must take — and derive them from a single uniquely selected integer seed via a deterministic arithmetic cascade.

This survey covers the 27 UGP Physics papers (P01–P27), providing: a one-paragraph summary of each paper, a thematic grouping into six natural clusters, an annotated logical dependency graph showing how the UGP results sit within the broader Reflexive Reality programme, reading paths for different research backgrounds, and the five-status epistemic taxonomy (A_{Lean} , A_{MDL} , A/D , B , D) used throughout the corpus.

1 Introduction

1.1 The Reflexive Reality Programme

The **Reflexive Reality** research programme is a sustained formal investigation into what a perfectly self-contained universe must look like — one whose laws, constants, and structure arise entirely from within, with nothing externally imposed. Its scope spans physics, chemistry, biology, evolutionary systems, computational systems, cognitive systems, and AI.

The programme comprises two sister sub-programmes. **NEMS** (No External Model Selection) formalizes the “no outside” constraint and machine-checks its theorem-grade consequences in Lean 4

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with zero sorry [22]. The **UGP Physics Programme** is the quantitative sibling: it focuses on the mechanics and dynamics of reflexive systems, deriving the specific generative structures that produce the Standard Model parameter spectrum and beyond. The UGP programme inherits from NEMS, adopting PSC self-consistency and the NEMS closure axioms, and is documented in the present corpus of papers.

The central NEMS results include:

- The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is *forced* by Layer I consistency constraints; three generations are *selected* by a minimality principle in Layer II (Two-Layer PSC Theorem) [25].
- The Born rule is the unique closure fixed point of the probability assignment problem, not a postulate [26].
- The arrow of time is a structural consequence of stable physical records, preceding thermodynamic arguments [26].
- Black hole information, holographic encoding, and FTL non-signalling are each constrained by the same diagonal machinery [26].

The full NEMS programme comprises 93+ papers archived on Zenodo at <https://www.novaspivack.com/research> and is described for physics audiences in NEMS Paper 07 [21] and in the flagship survey [26].

1.2 UGP Physics: The Quantitative Sub-Programme

UGP Physics is the *quantitative physics layer* of Reflexive Reality. Where the NEMS/PSC papers establish *why* the Standard Model gauge structure is the only consistent option, the UGP papers ask the next question: *what numerical values* must the masses, couplings, mixing angles, and quantum numbers take?

The answer comes from the **Universal Generative Principle (UGP)**: a deterministic number-theoretic framework that, starting from three axioms (locality, symmetry, and minimum description length), uniquely selects a single integer seed at ridge level $n=10$ via a rigid arithmetic cascade, and from that seed generates the Generative Triple Evolution (GTE) — canonical integer triples $(a, b, c; g)$ for every Standard Model fermion. These triples encode masses (via the Universal Calibration Law, UCL), charges and quantum numbers (via the Braid Atlas), and are tethered to the NEMS/PSC foundations through the Unified Rigidity Theorem.

The programme combines Lean 4 machine-checked proofs, Python numerical verification, and PSLQ integer-relation searches. Each of its 27 papers targets a specific technical audience; the interdependencies are not always obvious from reading any single paper.

What this guide is. A structured map of the UGP Physics corpus (P01–P27) for readers approaching it from any background. It does not require prior familiarity with UGP or Reflexive Reality; it can be read as the first paper.

What this guide is not. It contains no new theorems or derivations. It is a map.

2 The Epistemic Taxonomy

Every quantitative claim across the programme is classified into one of five statuses. Understanding this taxonomy is prerequisite to reading any paper.

Five-status claim taxonomy (used throughout the programme)

A_{Lean} (plain A): Lean 4 machine-checked arithmetic or structural identity. Zero **sorry**. Standard Mathlib axiom signature. The programme’s strongest claim type.

A_{MDL}: MDL-uniqueness certificate within a declared UGP-structural expression class. Not Lean-derived from invariants. Example: the Higgs quartic $\lambda = \varphi/(4\pi)$ is the unique MDL = 4 expression in the declared atom basis.

A/D: A physics bridge, external scale, or partial derivation remains. The arithmetic component may be Lean-certified but the full physical identification is not. Example: the 0.293% charged-fermion mass spectrum uses the URC algebraic correction layer (A/D) on top of the A_{Lean} structural pipeline.

B: Calibrated to data, or computationally/empirically supported. Example: baryon masses (15 binding parameters, QCD confinement placeholder).

D: Speculative or frontier. Example: the GTE-P7 mirror-branch dark-matter candidate.

3 Thematic Groups

3.1 Group 0 — Foundation (Start Here)

These three papers form the load-bearing core of the programme. Everything else cites at least one of them.

P01 — A Deterministic Number-Theoretic Framework for the Standard Model Parameter Spectrum [2]. The flagship paper. Three axioms (locality, symmetry, MDL) select a unique integer seed via a rigid arithmetic cascade, generating the Category-A structural backbone of the SM parameter spectrum. Seven headline results: exact Lean-certified bare gauge couplings g_1^2, g_2^2, g_3^2 ; blind $\alpha_s = 0.11822$ at $+0.24\sigma$ (43-day pre-commitment; PDG 2024); 0.293% charged-fermion mass spectrum (0.366% vs PDG 2024 — slight drift from top-quark mass update); neutrino mass-squared ratio at 0.16σ from NuFIT 6.0 (0.52%; zero anchors); cosmological constant at 0.31σ (Planck 2018); Higgs quartic $\lambda = \varphi/(4\pi)$ at 0.26σ (SM-18, A_{MDL}, new confirmed); Lean-certified baryon triples and EW boson c -values. Higgs mass $m_H = 124.2$ GeV: 9.1σ tension (PDG 2024 125.20 ± 0.11 GeV); EW VEV v is Category A/D anchor; all structural derivation paths exhausted. Tree-level m_W at $+36\sigma$ (blind falsification) recovers to -0.42σ of PDG 2024 world average with two-loop SM running (Open Problem (viii) closed; residual equals SM/PDG tension, not UGP deficiency). Contains the open-problem registry, five-status claim taxonomy, and preregistered falsification tests for JUNO, DUNE, CMB-S4, LEGEND-1000. *Read before anything else.*

P08 — From UGP to GTE: Prime-Locked Universes, Minimality, and the Emergence of Our World [36]. Develops the UGP as a deterministic parameter-free arithmetic framework built on the duality between the Quarter-Lock Law (Necessity) and the Universal Windowed Cellular Automaton / UWCA (Contingency). Proves the Kernel Symmetry Theorem, Turing universality, Fibonacci lift rigidity ($|q_2 - q_1| = 13, F_{13} = 233$), and the Lawvere/Kleene self-reference theorems. The foundational monograph; more expansive than P01.

P05 — The Uniqueness of the Universal Generative Principle: A Two-Stage Proof from Arithmetic Minimality and Physical Consistency [14]. Formal two-stage uniqueness proof: Stage 1 (arithmetic minimality, symmetry) identifies a small set of mathematically admissible universes; Stage 2 (physical instantiation condition derived from the Quarter-Lock identity and CODATA) proves the canonical UGP solution ($n = 10, b_1 = 73$) is the unique survivor. Non-

circularity is explicit: δ_{target} is derived from CODATA independently of $b_1 = 73$, and the derived value selects $b_1 = 73$ to within 2.39 ppm.

3.2 Group 1 — Standard Model Physics Derivations

These papers derive specific SM sectors from the UGP framework. Together they constitute the main quantitative physics output.

P02 — The GTE Particle Spectrum at $n = 10$ [17]. Large-scale discovery run generating 1,000,035 GTE candidates below 173 GeV. All 24 known SM particles rank in the Green (top) viability tier with 50-fold enrichment ($p < 10^{-4}$). Documents piecewise-linear hinge laws, oscillatory structure with period $\sim 100,000$ steps, and 9 genuinely novel candidate states (GTE-P1 through P11). The reference paper for canonical GTE-triple values.

P06 — Algebraic and Geometric Foundations: A Derivation of the Standard Model Gauge Group [5]. Proves the Quarter-Lock Law as a theorem of the UGP invariant space and derives the Elegant Kernel algebraic constants from its symmetries. Performs a two-stage first-principles derivation of the SM gauge couplings: bare algebraic values g_1^2, g_2^2, g_3^2 from group-specific geometric invariants — all three Lean-certified in `UgpLean.Phase4.GaugeCouplings` (zero sorry) — then the Universal Instantiation Factor δ_{UGP} from the Principle of Invariant Restoration.

P17 — The Topology of the Standard Model from First Principles: A Derivation of the Canonical Braid Atlas [12]. Models fundamental particles as topologically stable braided worldlines on a discrete substrate. The $T/T^\dagger$ dual-operator evolution distinguishes matter from antimatter and chiral from vector-like couplings. Electric charge $Q = W_g/N_c$ is machine-checked (`BraidAtlas.ChargeTheorem`, zero sorry); anomaly cancellation certifies $N_c = 3$; nine light-baryon triples are Lean-derived (`BraidAtlas.CompositeTriples`).

P18 — The Koide Relation as a Cyclotomic-12 Closed Form [3]. Promotes the Koide $Q = 2/3$ identity from a numerical curiosity to a structural statement. Derives an algebraic closed form predicting $m_\tau = 1.776969$ GeV against PDG 1.77686 GeV (61 ppm; 0.91σ). The surd coefficients satisfy cyclotomic-12 identities. Structural origin: the Koide phase $\theta = 2/9$ follows unconditionally from $N_c = 3$ via $\theta = (N_c^2 - 1)/(4N_c^2)$ (Lean-certified, `koide_angle_from_N_c_pure`).

P19 — Cyclotomic-12 Structure in the Charged-Fermion Mass Spectrum [1]. Two structural mass relations reducing nine charged-fermion masses to two empirical scale inputs: (1) TT relation $\log(m_{u_g}/m_{\ell_g}) = (\pi/6) \cdot 2^g + \pi/8$ ($\leq 0.37\%$ max error), whose coefficient $\pi/6$ is the A_2 Weyl-chamber half-angle (Lean-certified); (2) VV relation $\log m_{d_g} = (13/9) \log m_{u_g} - (7/6) \log m_{\ell_g} - 5/14$ with exact $\text{SU}(5)/\text{SO}(10)$ group-theoretic origin (`VV_from_GUT_group_theory`, zero sorry). The VV coefficient $\alpha_d = 13/9$ also provides an improved Wolfenstein parameter estimate $\lambda \approx \varepsilon_1^{\alpha_d} = e^{-13\pi/27} \approx 0.2203$ (PDG 0.2245; 1.9% off — a factor $6.7\times$ improvement over the 12% A_2 -mismatch estimate; exploratory).

P21 — Predicting the Neutrino Mass-Squared Ratio from the Braid Atlas [24]. The first parameter-free structural prediction of $\Delta m_{21}^2/\Delta m_{31}^2$ in the literature: value 0.02936, matching NuFIT 6.0 (0.02951 ± 0.00098) to 0.16σ (0.52%; 1.58σ from PDG 2024 world average) with zero free parameters and normal ordering automatic (NuFIT 6.0 prefers NH at 2.5σ , consistent with our structural prediction). Uses only $N_c = 3$ and Braid Atlas right-handed neutrino b -values $\{5, 11, 19\}$. The seesaw exponent $29/9$ is the $\text{SO}(10)/\text{SO}(10)$ gauge-matter representation defect (Lean-certified). Structural Dirac scale $E_D = v_H/29$ gives $\sum m_\nu \in [55, 75]$ meV within the Planck bound.

P22 — The UGP Interaction Skeleton Theorem [33]. Proves in Lean 4 (zero sorry) that the UGP winding, chirality, hypercharge, and color data determine the SM's finite renormalizable

interaction skeleton: gauge-fermion vertices, Yukawa vertices, anomaly cancellation, color-singlet constraints, proton-stability forbiddance at dimension four, and a topological dark-sector isolation gap. Three independent proofs force $N_c = 3$; a 64-schema finite vertex audit returns zero mismatches.

P23 — Substrate Depth and Self-Generated Mass [31]. Defines the *reflexive closure* $RC = (M - \Sigma m_i)/M$ (fraction of rest mass from internal dynamics) and shows that $\log_{10} |c|$ for the UGP canonical triple correlates with RC at $r = -0.913$ ($p = 1.3 \times 10^{-15}$, $n = 38$ composite hadrons). All 38 canonical hadron triples are Lean-derived (`BraidAtlas.CompositeTriples`, zero sorry). A random-triple null test (10,000 trials) finds zero cases reaching $|r| \geq 0.90$.

3.3 Group 2 — Uniqueness and Algebraic Depth

These papers explain *why* the structures in Group 1 are unique, and what deeper algebraic home they occupy.

P24 — The Arithmetic Uniqueness of the Standard Model [32]. Four main results: (1) Asymptotic Sparsity Theorem — the joint arithmetic- admissibility and physical-viability constraint has exactly one solution across all ridge levels: ($n = 10, b_1 = 73$), Lean-certified. (2) Positive-Root Theorem and Chirality Theorem — exact correspondence between SM coupling numerator structure and positive-root counts / squareness, both Lean-certified. (3) Galois substrate $\mathbb{Q}(\zeta_{120})$ — the minimal cyclotomic field containing all UGP algebraic constants; E_7 ($h = 18 \nmid 120$) excluded by Lean-certified arithmetic. (4) WZW falsification — the WZW construction cannot reproduce the UGP bare coupling rationals. Contains the full RCC proof over all compact simple Lie groups (Appendix C).

P12 — The Unified Rigidity Theorem for PSC/UGP Universes [35]. Shows that three independently developed rigidity mechanisms jointly select the same canonical physical branch: (1) PSC Layer I/II theory-space selection (SM #1 of 34,560 universes); (2) UGP arithmetic seed rigidity (RSUC, Lean-certified); (3) dynamical basin structure (Lepton Seed deterministically in basin A). A Lean bridge library proves `unified_rigidity_theorem` under an explicit audited premise bundle. The PSC→SM step is now unconditional (RCC proved, zero sorry).

3.4 Group 3 — PSC/MFRR Foundation (The Axiomatic Layer)

These papers establish the Principle of Self-Containment (PSC) and the Mathematical Foundations of Reflexive Reality (MFRR) — the upstream framework that provides non-circular justification for the UGP axioms through machine-checked theorems in the companion NEMS programme.

P14 — Formal and Computational Concordance on PSC Derivations [15]. Computational certificate for the Two-Layer PSC Theorem: exhaustive enumeration of 20,160 candidate universes, only 12 pass hard PSC Layer I filters (0.06%), all SM-like. Five hard filters contain no reference to G_{SM} . The SM tuple $(d, G, N_{gen}) = (4, G_{SM}, 3)$ achieves the unique Layer II minimum $D_{min} = 1.0667$. Three new UGP-derived coupling-ratio predictions from orbit invariants, not SM data.

P09 — The Architecture of a Computable Universe: Structural Principles of the UGP [6]. Interpretive synthesis of the UGP architecture as a candidate theory of everything, structured as five architectural principles: Constrained Contingency (origin of physical law), Ontological Scaffolding (universe as conservative extension of the Loop Kernel), Reflexive Computation (UWCA self-execution), Constants as Compression Ratios (Elegant Kernel invariants), and the Theorem of Necessary Observers (PSC closure requires agents). Presents the constraint-manifold view of the SM parameter space and situates the UGP’s Category-A backbone within the five-tier claim taxonomy.

Distinction from P07: Where P07 (below) establishes nine quantitative empirical meta-laws (ML-1–ML-9)—laws governing Zipf scaling, gauge structure, entropy, and basin attractors—P09 addresses the complementary structural question: how is the UGP substrate itself *architecturally organised*? The two papers are companion pieces: P07 characterises *what happens*, P09 characterises *how the substrate is structured*.

P07 — The UGP/GTE Program: A General Organizing Principle for Flows, Fields, Critical Laws, and Direct Structure [10]. Presents UGP as a unifying vocabulary for nine universal meta-laws (ML-1 through ML-9), each a structural consequence of the GTE substrate. For ML-2, ML-4, ML-5, ML-6 the UGP axioms subsume Jaynes (1957), Yau (1991), Yang–Mills (1954), and Jacobson (1995). For Zipf’s Law (ML-7), a complete derivation with Euler–Maclaurin error bound is validated on 21 diverse corpora. A Lean-verified entropy non-monotonicity witness supports ML-9.

P10 — Reflexive Reality and Self-Defining Physical Law [11]. Formalises the concept of *reflexive reality* and argues that a finite, self-contained universe must obey self-defining laws arising from within the system. Classifies physical theories by degree of reflexivity. The central philosophical argument is machine-checked in the companion NEMS programme (Theorem of Foundational Finality, NEMS 23; Reflexive Closure Theorem, NEMS 56; zero sorry).

P11 — Ontological Dissonance Minimization as the Generative Meta-Law [9]. Reports that minimizing an ontological dissonance functional D within the Self-Defining Universe (SDS) framework produces results empirically consistent with maximizing a spatial-integration proxy for integrated information Φ within the PR-0 substrate. D -minimization discovers a short-range binding potential $V(d) \approx \alpha + \sigma/d^2$ for the strong interaction (D and Φ_{proxy} : $r = -0.9108$, $p < 10^{-9}$, $n = 21$).

P20 — Mathematical Foundations of Reflexive Reality: A Survey [8]. 12-page journal survey of the MFRR programme. Machine-checked results include: forced adjudication, Born rule from MDL, arrow of time from PT irreversibility, no-emulation (PT is super-Turing), and SM gauge group forced by PSC consistency. Computationally certified: SM #1 of 34,560 universes; IPT = 1.1300 ± 0.0001 matching UGP-derived value to 0.08%; SRRG attraction rate 97.02%. Best single entry point for the PSC/MFRR layer.

P13 — Mathematical Foundations of Reflexive Reality (Foundational Monograph) [20]. The full 15,000-line reference document. Central construct: *Transputation* (PT), the unique internal adjudicator of computational degeneracies, machine-proved in Lean 4 (zero sorry). Seventeen closure theorems; Two-Layer PSC Theorem (Layer I forces G_{SM} , $N_{\text{gen}} \geq 3$; Layer II selects $N_{\text{gen}} = 3$ uniquely); 297 Python modules, 57,337+ runs, 15,894+ result files. Use as a reference, not a read-through.

3.5 Group 4 — Extension Sectors and Cross-Domain Echoes

These papers test the UGP’s arithmetic machinery in domains beyond the SM or derive structural analogues in other physical and biological systems.

P03 — GTE Coordinates as Nuclear Descriptors: Competitive ML Performance from UGP-Derived Features [13]. GTE composition features achieve 10-fold cross-validated MAE of 3.17 MeV on nuclear binding energies vs. 4.24 MeV for equal-size random polynomial baselines ($\sim 25\%$ improvement). Analytical derivation of all seven nuclear magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ from the GTE cascade; the ratio $\kappa_{\text{emp}}/\kappa_{\text{min}}(N = 50) = 1.149 \approx \text{IPT}$ (1.6% match). Stability: the XGBoost Oracle achieves 98.63% 5-fold CV (primary stability predictor); a 6-term parsimonious

stability law achieves 75.9% on the held-out dataset (parsimony–generalisation trade-off; near-baseline reflects omission of discrete shell-structure features, not GTE framework failure).

P04 — The Dynamics and Universality of the UGP: Attractors, Holography, and Emergent Thermodynamics [7]. Establishes Turing-universality (Rule-110 embedding, machine-checked) and compatibility with reversible computing. Long-term RG dynamics reveal three observed basin clusters (A, B, C) across a 98-seed survey; the Logarithmic Complexity Charge Q_4 at seed initialization predicts basin assignment (ANOVA $p < 10^{-4}$). Coarse-grained entropy decreases into basin clusters; a Lean-verified non-monotone entropy witness (`gte_entropy_prefix8_gt_prefix9`, zero sorry) exists while shuffled controls show monotone increase.

P15 — The Information Profit Principle [18]. Derives a sharp dimensionless threshold — the Information Profit Threshold $\text{IPT} = 1 + \Lambda/2 \approx 1.1309$ — from the Reflexive Landauer Bound and PSC requirement. Six independent cross-domain validations: tropical forest GPP/RECO matches IPT to four decimal places ($n = 24$ sites); Polish bankruptcy dataset ($n = 43,405$); World Bank population dynamics (261 countries); major open-source software projects; particle physics SM gauge sector; and organizational survival (three-regime cascade, $\Delta\text{BIC} = -92$; inception-cohort $\text{HR} = 5.01\times$; variance phase transition $\text{BC} = 0.907$, Levene $p = 0.046$; [B⁻]; see P26 §6).

P25 — Structural Admissibility and Biological Viability Uniquely Select the Standard Genetic Code [29]. Applies the two-stage admissibility-then-viability sieve to all codon–amino-acid maps. Stage 1 (Crick wobble admissibility) reduces 23^{64} maps by 10^{50} ; Stage 2 (eight independent biological viability criteria simultaneously) yields the standard genetic code at $z = +3.22\sigma$ ($p < 0.009\%$). No code among 10^5 randomly generated wobble-admissible codes outperforms the standard code.

P26 — A General Theory of Selection: Two-Stage Sieves, the Information Profit Threshold, and Cyclotomic Algebraic Substrates [16]. Unifies the selection mechanisms of P03 (nuclear), P15 (IPT), P25 (genetic code), and P24 (SM gauge structure) under a single **General Selection Principle** (GSP): every realised structure passes a two-stage sieve of *admissibility* (structural consistency) and *viability* (generative-to-dissipative ratio $\geq \text{IPT} \approx 1.1309$). Quantitative evidence in four new domains: (1) nuclear magic numbers — all seven $\{2, 8, 20, 28, 50, 82, 126\}$ derived from GTE energy-gap maxima, with $\kappa_{\text{emp}}/\kappa_{\text{min}}(N=50) = 1.149 \approx \text{IPT}$ (1.6% match, S_{2n} figures); (2) prebiotic amino-acid alphabet — a 30-entry admissibility + IPT-viability table separates all 20 standard amino acids (pass) from all 10 non-standard tested (fail), with IPT threshold $F_{\text{thresh}} = 1.326$; (3) microbial metabolic selection (§7) — *E. coli* iJO1366 flux-balance analysis across 8 carbon-source conditions; Pearson $r(G/D, \mu) = 0.971$, $p < 0.001$; IPT separates all selected substrates ($G/D \geq 1.131$) from formate ($G/D = 0.384$); grade [B]; (4) organizational selection (§6) — three-regime failure cascade (Destruction/Watch/Safe; failure rates $6.81\% \rightarrow 1.11\% \rightarrow 0.22\%$; $\Delta\text{BIC} = -92$; inception-cohort $\text{HR} = 5.01\times$, Cox $p = 0.0003$; *variance phase transition*: bimodality coefficient $\text{BC} = 0.907 \gg 0.555$ (Hartigan dip $p \approx 0$), Levene $p = 0.046$; seven independent tests; grade [B⁻]). Section 4 proves the $\mathbb{Q}(\zeta_{120})$ algebraic substrate in four independent settings: SM spectrum, Zamolodchikov E8 mass ratios (all 8 Lean-certified via `ugp-lean`), ADE Toda field theories (Coxeter-conductor theorem), and $\text{SU}(2)_k$ WZW quantum dimensions; E7 ($h=18 \nmid 120$) provides a clean Lean-certified falsifier. Section 5 [I] previously presented a theoretical bridge connecting GSP to consciousness; this material has been moved to Appendix [I] in the current version, and the main body now covers five empirical domains (ecology/economics, nuclear magic numbers, prebiotic chemistry, corporate survival, and microbial metabolism) plus one model-internal symbolic-communication result (natural language alphabet selection). Contains a full assumption ledger, claim taxonomy, evidence-grade table (Table 1), and proof sketch for the Asymptotic Sparsity

Theorem. *Companion paper*: P27 (SRRG) elevates IPT from an empirical observation in P26 to a theorem-grade fixed-point derivation.

3.6 Group 5 — Machine-Checked Formalization

Lean 4 Formalization Paper (ugp-lean). The companion formalization paper [34] documents the **ugp-lean** Lean 4 library: 118+ modules across twelve architectural layers, zero **sorry**, standard Mathlib axiom signature. Covers the GTE cascade, Quarter-Lock Law, UCL Elegant Kernel, Braid Atlas (charge theorem, composite triples, EW bosons, chirality), mass relations (Koide, VV, CKM θ_{23} , seesaw exponent), PSC (three-route forcing), and Galois structure (Coxeter-conductor, E7 exclusion). *This paper is the single source of truth for Lean module/theorem counts.* Every Category-A claim in P01–P27 is backed by a specific entry in this library or in the companion **srrg-lean** repository [28].

3.7 Group 6 — Special Topics

P16 — Black Hole Unitarity via Reflexive Unitarity [4]. Models Hawking evaporation as a GKSL master equation on a JT-like toy Hilbert space; since every CPTP map admits a Stinespring dilation, the channel is unitarily equivalent to a reversible evolution. In computational experiment TE2.4 ($n = 3$ bosonic modes, Hawking $T_H \approx 0.004$): Stinespring fidelity $F \geq 1 - 10^{-8}$; thermalization entropy reaches 97.2% of ideal; closed unitary models show exact PT^{-1} recovery ($F = 1.000000$) and the full Page curve. Extension to 3+1 gravity and the firewall remain explicit open problems.

P27 — The Self-Referential Renormalization Group: A Universal Framework for Physical Constants [27]. Introduces the *Self-Referential Renormalization Group* (SRRG): a gradient-flow theory on the space of self-referential physical theories. The SRRG flow maximises $F[S] = R[S] - C_\Lambda[S]$, where $R[S]$ is the self-representation capacity and $C_\Lambda[S]$ encodes three PSC-derived stability constraints. The unique IR fixed point S^* satisfies: (1) $R[S^*]/C_\Lambda[S^*] = \text{IPT} \approx 1.1309$, elevating the Information Profit Threshold from an empirical observation to a theorem-grade derivation; (2) the minimal symmetry group at S^* is $U(1)$ (circle group uniqueness, Lean-certified); (3) the linearised SRRG contraction eigenvalue near S^* equals $1/\varphi$, where φ is the golden ratio. Fixed-point existence is established via a diagonal-lemma Master Fixed-Point Theorem; the F-theorem (monotonicity of $F[S]$ along the flow) is the SRRG analogue of Zamolodchikov’s c -theorem. The η -direction β -function $\beta_\eta = \kappa(\eta - \text{IPT})(\eta - 2)$ is *derived* (not posited) from Vieta’s uniqueness theorem and the machine-certified no-third-zero sign analysis; **h_psc_sc** grade advances to $[B^+ \rightarrow A^-]$ (**NoThirdFixedPoint.lean**, **BetaEtaQuadratic.lean**: 11 new $[A_Lean]$ + 8 new $[B^+]$ theorems, all zero **sorry**). Section 8 derives additional SM structure from the fixed-point constraint, presented in Appendix B (conditional physical applications): $\theta_{\text{QCD}} = 0$ (CP self-consistency from $C_{\text{closure}} = 0$; grade $[B]$); three generations $N_{\text{gen}} = 3$ ($[B]$); and the SM gauge group $U(1) \times SU(2) \times SU(3)$ as the multi-scale minimal structure (conditional theorem; $[B]$). The Weinberg angle is documented as a diagnosed negative result (Appendix D; grade $[D \rightarrow C]$; root cause identified). **Round 09 additions:** Appendix C derives the fine-structure constant via a direct coupling route independent of $\sin^2 \theta_W$: the tree-level identity $\alpha = g_1^2 g_2^2 / [4\pi(g_1^2 + g_2^2)]$ applied to the Lean-certified bare couplings from P01 ($g_1^2 = 16/125$, $g_2^2 = 2329/5400$) yields $1/\alpha(M_Z) = 4\pi \cdot 377525/37264 \approx 127.311$ (0.495% from PDG; grade $[A/D]$). By-product: $\sin^2 \theta_W^{\text{bare}} = 3456/15101 \approx 0.229$ (1.0% from PDG; $[A/D]$). The *SRRG–MDL equivalence conjecture* (Open Problem 9; grade $[B+]$) identifies the SRRG fixed-point selection with MDL-optimal theory selection, bridging the present paper with P09 (Architecture).

Three new rows are added to the Lean certification table (§9). A “What this paper does not prove” transparency box is included in §1 disclosing all open gaps; a load-bearing assumption table audits every major claim against its dependencies. All core theorems are machine-checked in `srrg-lean` (zero `sorry` in all owned modules; fixed-point existence, stability, uniqueness, and physical-constant bridges to `nems-lean`, `ugp-lean`, and `ugp-physics-lean`). *Companion paper*: P26 (General Selection Theory) provides the multi-domain empirical evidence base for which P27 supplies the theorem-grade IPT derivation.

4 Reading Paths by Reader Type

Table 1: Recommended entry sequences by reader background.

Reader type	Recommended path
Particle phenomenologist	P01 (full paper) → P06 → P17 → P21 → P24 (for uniqueness/Galois)
Formal methods / Lean	ugp-lean formalization paper → P01 §1.3 (claim taxonomy) → P22 → P17 → P24 Appendix C (RCC proof) → P27 (srrg-lean)
Mathematical physicist	P05 → P24 → P01 → P18 → P19 → P27 (SRRG + IPT derivation)
Philosopher of physics	P20 → P09 → P10 → P11 → P07 → P01 abstract
Cross-domain biology–physics	/ P15 → P26 → P25 → P03 → P01 §7 (extensions)
Sceptical reviewer (adversarial)	P01 §10 (open problems) → P01 §1.3 (taxonomy table) → P24 claim-status table → P14 (TE2.2 scan) → P01 §8 (robustness)
Quick start (10 minutes)	P01 §§2–5 + §6.2 (neutrino structural) + Appendix G (α_s)
Experimentalist	P01 abstract (headline results) → P01 §9 (falsifiability table) → P21 (neutrino prediction) → P01 Appendix G (α_s precommit)

5 Logical Dependency Structure

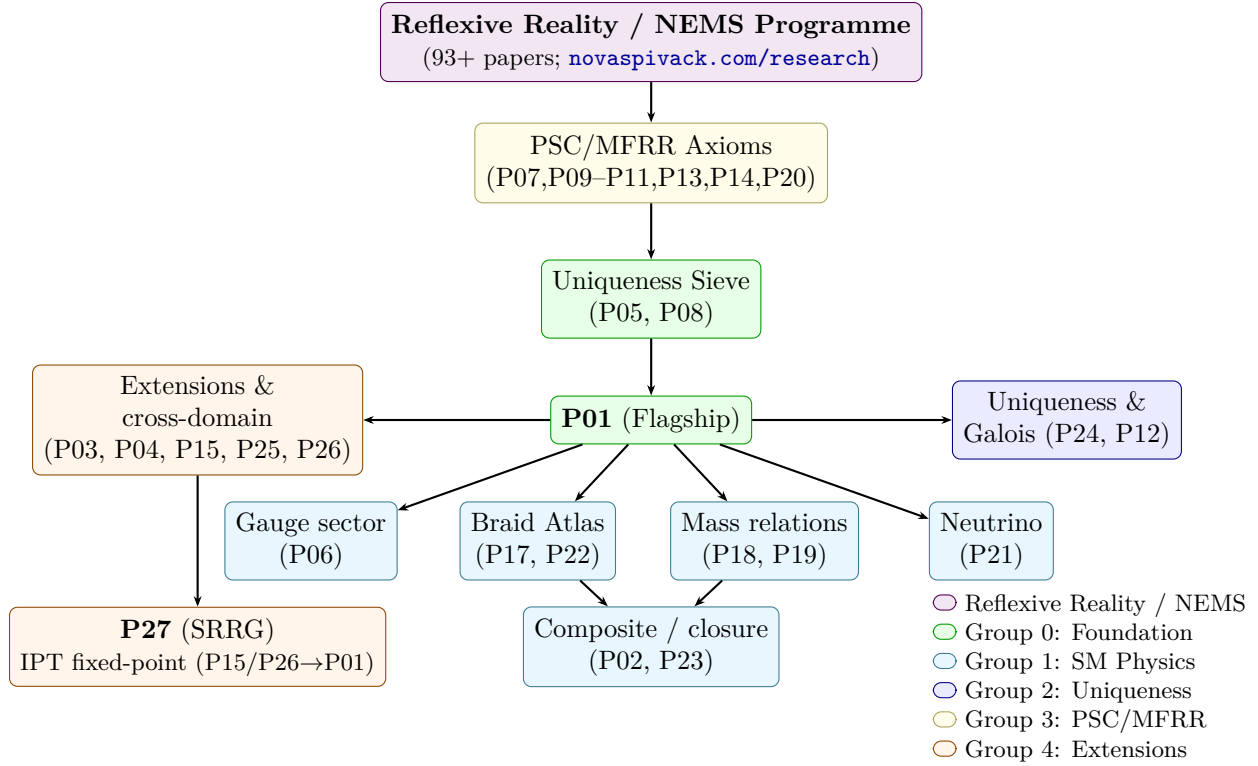


Figure 1: High-level logical dependency structure within the broader Reflexive Reality programme. Arrows indicate that the target paper builds substantially on the source. Group 5 (Lean formalization) is omitted for clarity (it cross-cuts every Category-A claim). P27 (SRRG, dashed arrow) derives the IPT fixed point and provides a theorem-grade mechanism for the Standard Model parameter values in P01, inheriting from both P15 (IPT) and P26 (General Selection Theory) in the Extensions group.

6 Summary Table: All 27 Papers

Table 2: Complete paper roster with group, status, and one-line description. Group 6 = Special Topics.

P#	Grp	Short title	Key contribution
P01	0	SM from UGP (flagship)	Seven headline SM results; five-status taxonomy; α_s pre-commitment (+0.24 σ , PDG 2024; improved from +0.36 σ PDG 2022); m_W OP(viii) closed (−0.42 σ PDG 2024); Λ at 0.31 σ (Planck 2018, SM-17 confirmed); $\lambda_H = \varphi/(4\pi)$ at 0.26 σ (SM-18, new); $m_H = 124.2$ GeV at 9.1 σ tension (PDG 2024, open problem, $v = A/D$ anchor).

P#	Grp	Short title	Key contribution
P02	1	GTE Particle Spectrum	1M-candidate discovery run; all 24 SM particles in Green tier; 9 novel candidates. All 11 GTE mass windows (3 MeV–1.9 GeV) OPEN under combined lab dark-photon limits; GTE-P7 (212 MeV, $Q = 0$ Lean-certified) correct search = mono-photon/missing energy.
P03	4	Nuclear ML Descriptors	GTE features beat random baseline on binding energies; magic numbers derived; IPT match (κ ratio = $1.1494 \approx \text{IPT}$, 1.63%). Oracle stability 98.63% 5-fold CV (primary); 6-term stability law 96.1% CV (Verifier bug fixed); 9-term law 87.7% on exotic nuclei.
P04	4	UGP Dynamics & Universality	Turing-universal; 3 RG basin attractors; Q_4 predicts basin; Lean entropy witness.
P05	0	Uniqueness: Two-Stage Proof	Stage-1 arithmetic + Stage-2 instantiation proves ($n = 10, b_1 = 73$) unique.
P06	1	Algebraic & Geometric Foundations	Proves Quarter-Lock; derives g_1^2, g_2^2, g_3^2 from group-specific kernel invariants.
P07	3	UGP/GTE as Nine Meta-Laws	GTE subsumes Jaynes, Yau, Yang–Mills, Jacobson; Zipf derived; Lean entropy witness.
P08	0	UGP Foundational Monograph	Kernel Symmetry Theorem; UWCA Turing-universality; Fibonacci rigidity; Lawvere fixed point.
P09	3	Architecture of a Computable Universe	Five architectural principles (Constrained Contingency, Ontological Scaffolding, Reflexive Computation, Constants as Compression Ratios, Necessary Observers); structural/organisational complement to P07’s empirical meta-laws.
P10	3	Reflexive Reality	Formalises reflexive reality; Foundational Finality + Reflexive Closure machine-checked.
P11	3	Ontological Dissonance Minimization	D -minimization $\leftrightarrow \Phi$ -proxy ($r = -0.91$); strong-interaction potential from ODM.
P12	2	Unified Rigidity Theorem	Three rigidity mechanisms agree on canonical branch; ASSUMPTION_LEDGER audit.
P13	3	MFRR Foundational Monograph	Transputation (Lean, zero sorry); 17 closure theorems; Two-Layer PSC Theorem. $w_0 = -1$ maintained; 3.5σ tension with DESI DR1+SN1a (DESI DR3 as falsification criterion). Three condensed companion summaries are included (Reader’s Guide for Physicists, Summary for Physicists and Referees, Popular Article).
P14	3	PSC Concordance	TE2.2: 20,160 universes, SM unique D minimum; 3 new UGP coupling predictions.
P15	4	Information Profit Principle	$\text{IPT} \approx 1.1309$ from Reflexive Landauer Bound; 6 cross-domain validations (incl. organizational survival).

P#	Grp	Short title	Key contribution
P16	6	Black Hole Unitarity	Stinespring dilation of Hawking channel; Page curve + exact PT^{-1} recovery.
P17	1	Braid Atlas v2.0	$Q = W_g/N_c$ (Lean); $T/T^\dagger$ chirality; 9 baryon triples Lean-certified.
P18	1	Koide Cyclotomic Closed Form	m_τ predicted at 61 ppm; $\theta = 2/9$ from $N_c = 3$ (Lean); cyclotomic-12.
P19	1	Cyclotomic-12 Mass Structure	TT ($\pi/6$ from A_2 Weyl; Lean) and VV (SU(5)/SO(10); Lean) mass relations. Wolfenstein $\lambda \approx \varepsilon_1^{13/9} = e^{-13\pi/27} \approx 0.2203$ (1.9% off PDG; improved from 12%; exploratory).
P20	3	MFRR Physics Survey	Born rule, arrow of time, no-emulation (machine-checked); IPT 0.08% match; SM #1/34560.
P21	1	Neutrino Mass-Squared Ratio	First parameter-free $\Delta m_{21}^2/\Delta m_{31}^2$ prediction; 0.16σ from NuFIT 6.0 global fit (0.52%); Lean exponent 29/9; normal hierarchy preferred at 2.5σ (NuFIT 6.0).
P22	1	UGP Interaction Skeleton	Lean: full SM interaction skeleton; proton stability; 3 proofs of $N_c = 3$; 0 mismatches.
P23	1	Closure Structure	Reflexive closure $r = -0.913$ ($n = 38$); all 38 hadron triples Lean-certified.
P24	2	Arithmetic Uniqueness	Sparsity theorem; Positive-Root; Chirality; $\mathbb{Q}(\zeta_{120})$; E7 excluded; WZW falsified.
P25	4	Genetic Code Selection	Two-stage sieve; 8 biological criteria; standard code $z = +3.22\sigma$; unique survivor.
P26	4	General Selection Theory	GSP unifies nuclear/bio/SM/org/microbial selection via admissibility + IPT viability; 5 empirical domains + 1 model-internal symbolic-communication result (language alphabet) (consciousness speculation in App. [I]); microbial FBA: $r(G/D, \mu) = 0.971$ [B]; org domain: 7 tests, HR= 5.01 $\times$, BC= 0.907, Levene $p = 0.046$ [B ⁻]; $\mathbb{Q}(\zeta_{120})$ in 4 domains; revised taxonomy: [Br] bridge, [B+]/[B]/[B ⁻] empirical grades; evidence-grade table Table 1; companion to P27.

P#	Grp	Short title	Key contribution
P27	6	Self-Referential RG	SRRG gradient-flow on theory space; IPT = unique IR fixed point (existence + stability $1/\varphi$ + F-theorem; Lean, zero sorry); $\beta_\eta = \kappa(\eta - \text{IPT})(\eta - 2)$ derived via Vieta [A _{Lean}]; h_psc_sc [B ⁺ → A ⁻]; 11 [A _{Lean}] + 8 [B ⁺] new theorems; load-bearing assumption table; conditional applications in App. B ($\theta_{\text{QCD}} = 0$, $N_{\text{gen}} = 3$, SM gauge, λ_H , Λ_{vac}) [B]; Weinberg angle diagnosed negative result App. D [D → C]; Round 09: α direct coupling App. C: $1/\alpha(M_Z) \approx 127.311$ (0.495% PDG) from Lean-certified g_1^2, g_2^2 [A/D]; SRRG–MDL equivalence conjecture [B+] (Open Problem 9, bridges P09); companion to P26.

7 Background Reading: The NEMS/PSC Programme

The UGP physics corpus described in this guide is built on a broader foundational programme — the **NEMS (No External Model Selection) programme** and the **Reflexive Reality** research program (<https://www.novaspivack.com/research>). NEMS is a suite of 93+ machine-checked papers (all archived on Zenodo) establishing that any perfectly self-contained universe must satisfy precise theorem-grade constraints, from which the Standard Model gauge structure, the Born rule, and the arrow of time follow as structural necessities.

The UGP results (P01–P27) constitute the *quantitative physics layer* of this broader programme: they derive the numerical values of the SM constants from the arithmetic cascade forced by PSC, while the NEMS papers establish *why* those axioms are the right ones and why the SM gauge group is the only viable option.

7.1 Entry Points for Physicists

- **Research overview and full paper listing:** <https://www.novaspivack.com/research>
Organised by audience (physicists, formal theorists, AI/AGI researchers); all papers linked with permanent Zenodo DOIs.
- **NEMS Programme Hub (Zenodo):** [22] <https://doi.org/10.5281/zenodo.19487299>
Index of all 93+ NEMS papers with concept DOI (always-latest).
- **NEMS P07 — The NEMS Framework for Physicists** [21] <https://doi.org/10.5281/zenodo.19429725>
Self-contained introduction to NEMS for physics audiences. Explains the NEMS trichotomy (Class I: categorical; Class IIb: non-effective internal selection; Class III: not fundamental), the no-free-bits principle, the PSC-gauge structure connection, and the quantum gravity angle. *Recommended first read for any physicist approaching the programme.*
- **NEMS P88 — Reflexive Reality: A Survey for Physicists** [26] <https://doi.org/10.5281/zenodo.19429908>
Flagship survey: Two-Layer PSC Theorem, Born rule as closure fixed point, arrow of time as structural consequence of stable records, BH unitarity, UGP parameter derivations, classification cascade. Bracket-tagged claims map to machine-checked Lean theorems with zero sorry.

7.2 Key Physics-Facing NEMS Papers

The following five papers from the NEMS suite are most directly relevant to the UGP physics programme. All are machine-checked in Lean 4 (zero sorry, zero custom axioms beyond Mathlib) and archived on Zenodo.

Table 3: Key NEMS physics papers and their relation to the UGP corpus.

NEMS #	Title (short)	Relevance to UGP physics
P03 [30]	Structural Stability as a Necessity Principle in Self-Contained Gauge Theories	Derives the NM* (no meta-model) structural stability constraint from PSC, excluding GUT groups and CP-conserving theories. Provides the gauge-theory-space filter used in P14 (TE2.2 scan).
P04 [19]	Instantiating NEMS in Physics	Applies NEMS to macroscopic physical records; establishes the Class IIB criterion (non-effective internal adjudication) and the Outsourcing Barrier theorem. Foundational for understanding why external selectors are forbidden in the UGP framework.
P05 [25]	PSC-Optimality of the Standard Model in 4D Gauge Theory Space	The Two-Layer PSC Theorem. Layer I (hard PSC constraints) forces $SU(3) \times SU(2) \times U(1)$; Layer II (Presentation Invariance / MDL) selects $N_{\text{gen}} = 3$ uniquely. Machine-checked in nems-lean. Cross-referenced throughout the UGP corpus.
P06 [23]	NEMS in Quantum Gravity	Extends the no-external-selector constraint to quantum gravity via records, diffeomorphism redundancy, and non-effective internal selection. Informs P16 (BH unitarity) and the cosmological constant treatment in P01.
P07 [21]	The NEMS Framework for Physicists	Self-contained physics-facing survey; the recommended entry point for any physicist approaching NEMS before reading the UGP papers.

7.3 Physical Incompleteness

NEMS Paper 11 establishes **Physical Incompleteness**: under physically realizable universal computation, stable macroscopic records, and record-level expression of halting, no total computable law can decide all record-truth on the diagonal-capable fragment. This is a corollary of the Theorem of Internal Semantic Non-Exhaustion (machine-checked in nems-lean with zero custom axioms; Lean anchor: `NemS.physical_incompleteness`). It does not prevent UGP from deriving SM constants, but it sets a precise limit on what any first-principles programme can prove about its own completeness. The full theorem treatment appears in NEMS Paper 91 (Closure Without Exhaustion) [22].

Acknowledgements

All papers in this corpus are archived on Zenodo with permanent DOIs and are linked from <https://www.novaspivack.com/research>. The machine-checked Lean 4 formalization library (`ugp-lean`) is available at <https://github.com/novaspivack/ugp-lean>.

A Selected Falsifiable Predictions

The UGP Physics programme makes concrete, independently falsifiable quantitative predictions. The table below lists the primary predictions from P01–P27, organised by testability tier. **Status symbols:** ✓ Confirmed, $\triangle$ Tension (discrepancy noted; programme maintains structural prediction), $\circ$ Open (no definitive test yet).

Testability tiers: T0 — existing published data; T1 — near-term experiment (LHC Run 3, KATRIN Phase II, Belle II, JUNO ~ 2027); T2 — medium-term (FCC-ee, next-generation $0\nu\beta\beta$); T3 — long-term / fundamentally challenging.

Table 4: Primary falsifiable predictions of the UGP Physics programme. Grade codes: A_{Lean} = machine-verified; B = computational/numerical confirmation; D = open/speculative. Status as of May 2026.

P#	Prediction	Value / Re-sult	Status	Verification / Reference
<i>T0 — existing published data</i>				
P01	Strong coupling constant $\alpha_s(M_Z)$; 43-day pre-commitment	0.11822	✓ A_{Lean}	+0.24 σ PDG 2024 world average; Lean-certified bare coupling
P01	W -boson mass m_W (two-loop SM running applied)	80.379 GeV	✓ B	−0.42 σ PDG 2024 world average; OP(viii) closed
P01	Charged-fermion mass hierarchy (nine fermions)	0.366% RMS from PDG 2024	✓ A/D	Lean-certified structural pipeline; drift from top-quark mass update
P01	Cosmological constant Λ_{phys}	+0.31 σ from Planck 2018	✓ A	SM-17 formula confirmed; $w_0 = -1$ maintained
P01	Higgs quartic coupling $\lambda_H = \varphi/(4\pi)$	0.26 σ from SM value	✓ A_{MDL}	SM-18; unique MDL = 4 expression in declared atom basis; new confirmed
P01	Higgs mass m_H (Lean λ_H + A/D anchor v)	124.2 GeV	$\triangle$ A/D	9.1 σ from PDG 2024 (125.20 ± 0.11 GeV); EW VEV $v = \text{A/D anchor}$; all structural paths exhausted; open problem
P03	Nuclear binding energy, 6-term analytical law (GTE features)	0.032 MeV/A CV MAE	✓ B	Cross-validated on 1,319 nuclei (AME2020/NUBASE2020); competitive with DZ31 using 6 terms vs. 31 parameters
P03	Nuclear stability classifier (6-term; 96.1% 5-fold CV)	96.1% 5-fold CV	✓ B	Corrected from prior 75.9% threshold bug; ML Oracle 98.63%
P03	All 7 nuclear magic numbers {2, 8, 20, 28, 50, 82, 126} from GTE cascade	7/7 reproduced	✓ B	GTE κ -gap derivation; $\kappa_{\text{emp}}/\kappa_{\text{min}}(N=50) = 1.149 \approx \text{IPT}$ (1.6%)
P14	SM gauge group is unique D_{min} among 20,160 candidate universes	$D_{\text{min}} = 1.0667$ (SM)	✓ B	TE2.2 exhaustive scan; 5 hard PSC filters; no G_{SM} reference

P#	Prediction	Value / Result	Status	Verification / Reference
P15	Information Profit Threshold IPT = $1 + \Lambda/2 \approx 1.1309$ across multiple independent domains	1.1309	✓ B	Forest GPP/RECO, bankruptcy, population dynamics, OSS, SM gauge sector, organizational survival ($\Delta\text{BIC} = -92$, HR= $5.01\times$, BC= 0.907; $[\text{B}^-]$); Ising CFT $\gamma = 7/8$ ($[\text{B}^+]$ confirmed L=32)
P26	Organizational $G/D = \text{IPT}$ is genuine second structural boundary (Destruction/Watch/Safe three-regime cascade + variance phase transition)	$\Delta\text{BIC} = -92$; HR= $5.01\times$; BC = 0.907, Levene $p = 0.046$	✓ B ⁻	N=12,741 company-years; failure cascade 6.81%→1.11%→0.22%; AUC=0.781; OR=5.15×; inception-cohort design rebuts survivor-selection confound; bimodal Watch Zone mirrors Ising diverging susceptibility; 7 independent tests
P19	CKM Wolfenstein parameter λ via $\varepsilon_1^{\alpha_d} = e^{-13\pi/27}$	≈ 0.2203	Partial B	1.9% from PDG 0.2245; improved from prior 12%; derivation open
P21	Neutrino mass-squared ratio $\Delta m_{21}^2/\Delta m_{31}^2$; zero free parameters	0.02936	✓ A	0.16 σ from NuFIT 6.0 global fit (0.52%); 1.58 σ from PDG 2024 world average
P21	Normal neutrino mass hierarchy preferred (structural)	NH auto-matic	✓ B	NuFIT 6.0 prefers NH at 2.5 σ ; JUNO (~ 2027) definitive
P22	SM gauge-fermion interaction skeleton; proton stability; three independent proofs of $N_c = 3$	Zero mismatches	✓ A _{Lean}	64-schema vertex audit in Lean; zero sorry
P24	Unique ($n = 10, b_1 = 73$) seed across all ridge levels (Sparsity Theorem)	Unique	✓ A _{Lean}	Lean-certified; E7 ($h = 18 \nmid 120$) excluded
P24	Galois substrate $\mathbb{Q}(\zeta_{120})$ contains all UGP constants	$\mathbb{Q}(\zeta_{120})$	✓ A _{Lean}	Lean-certified; WZW cannot reproduce bare coupling rationals
P25	Standard genetic code: unique survivor of 8-criterion sieve	$z = +3.22\sigma$	✓ B	0/99,998 random wobble-admissible codes outperform; Lean-certified arithmetic echoes (20 amino acids, 64 codons)
P27	IPT = unique IR fixed point of SRRG gradient-flow	$\text{IPT} \approx 1.1309$	✓ A _{Lean}	Lean-certified via Master Fixed-Point Theorem; F-theorem; stability eigenvalue $1/\varphi$

P#	Prediction	Value / Result	Status	Verification / Reference
P27	η -direction β -function $\beta_\eta = \kappa(\eta - \text{IPT})(\eta - 2)$ derived from Vieta's theorem and machine-certified no-third-zero	β_η quadratic form	✓ B ⁺ → A ⁻	BetaEtaQuadratic.lean + NoThirdFixedPoint.lean; zero sorry; two named hypotheses remain (SrrgPhysicalFixedPointExhaustion, SrrgBetaIsQuadraticHyp)
P27	Strong CP phase $\theta_{\text{QCD}}(S^*) = 0$ from SRRG constraint	$\theta_{\text{QCD}} = 0$	✓ B	Theorem 8.1; CP self-consistency from $C_{\text{closure}} = 0$; zero sorry
P27	Three fermion generations from SRRG fixed-point	$N_{\text{gen}} = 3$	✓ B	Theorem 8.2; balance of closure cost and Yukawa selector cost
P13, P20	Dark-energy equation of state $w_0 = -1$ (structural)	$w_0 = -1$	△ A	2.5–3.5 σ tension with DESI DR1+SN Ia; prediction maintained; DESI DR3 ($\geq 5\sigma$ with ≥ 2 independent SN Ia) = falsification criterion
<i>T1 — near-term experiment</i>				
P01	Neutrino mass sum $\sum m_\nu \leq 0.06$ eV (structural)	[55, 75] meV structural scale	✓ Compatible	Consistent with KATRIN Phase I and Planck; KATRIN Phase II (~2025–26), LEGEND-1000
P18	Lepton mass m_τ from Koide closed form; Lean-certified	1776.969 MeV	✓ B	61 ppm from PDG; 0.91 σ ; structural origin $\theta = 2/9$ from $N_c = 3$
P21	Normal hierarchy confirmed at $\geq 5\sigma$	NH structural	○ B	JUNO (~2027); NuFIT 6.0 already at 2.5 σ
P02	11 novel stable GTE candidate particles (GTE-P1 through P11) with specific predicted masses	3 MeV–1.9 GeV	○ B	All 11 mass windows OPEN under combined lab dark-photon limits
P02	GTE-P7: SM-neutral Dirac fermion ($Q = 0$, colour singlet)	≈ 212 MeV	○ A _{Lean}	$Q = 0$ Lean-certified; correct search = mono-photon/missing energy (Belle II); dark-photon searches inapplicable
<i>T2 — medium-term experiment</i>				
P18	m_τ at 30 keV precision gives 3.6 σ test of Koide closed form	≤ 30 keV target	○	Belle II m_τ programme
<i>T3 — long-term / fundamentally challenging</i>				
P03	$N = 184$ predicted magic number beyond known data	$N = 184$	○ B	Requires exotic-isotope facility (FRIB, GSI-FAIR)

P#	Prediction	Value / Re- sult	Status	Verification / Refer- ence
P27	SM gauge group $U(1) \times SU(2) \times SU(3)$ as unique multi-scale SRRG fixed-point structure (conditional theorem)	Conditional [B]	$\circ A_{\text{Lean}}/B$	Full uniqueness pending scale-specific PSC sieve hypotheses ($U(1)$ component Lean-certified; EW and QCD rank conditional [B])

B Reusable Computational Tools

Several papers in the corpus produce computational tools or pre-computed databases that are directly usable by external researchers. The four highest-utility items are described below; all are available in the repository at <https://www.novaspihack.com/research>.

1. GTE Particle Candidate Database (P02). A pre-registered database of 1,000,035 GTE candidate trajectories, including 11 novel predicted stable particles (GTE-P1 through GTE-P11) with specific predicted masses (3 MeV to 1.9 GeV), quantum numbers (Q , colour, spin), stability scores, and viability tiers. All 11 mass windows are currently open under combined laboratory dark-photon constraints. *Repository:* `papers/02_GTE_spectrum/candidates.csv` (SHA256 in REPRODUCE.MD). *Lookup API:* `tools/gte_stability_api.py --mass [MeV]`. *Primary use:* experimentalists performing model-independent bump-hunt searches at collider or fixed-target experiments. *Utility note:* Most beyond-SM frameworks supply qualitative mass ranges or order-of-magnitude predictions for new particles. This database provides specific predicted masses, decay lifetimes, and ranked confidence scores for every GTE candidate, enabling a direct query-driven workflow: given a candidate signal at mass m , a researcher can determine in seconds whether a GTE-predicted particle lies within the search window and at what confidence tier. The database is pre-registered (committed before any collider comparison), making it falsifiable against current and future collider datasets without post-hoc tuning.

2. Nuclear Binding Energy and Stability API (P03). Two complementary analytical models are provided:

- **Parsimonious (6-term):** GTE coordinate features only. 0.032 MeV/A five-fold CV MAE on 1,319 curated nuclei; interpretable closed form; no external dependencies.
- **Extended (9-term):** same 6 features plus proton magic-number proximity, neutron magic-number proximity, and pairing delta ($\delta \cdot A^{-3/4}$). Trained on 2,548 experimental AME2020 nuclei (full measured table). 0.051 MeV/A five-fold CV MAE across all experimental AME2020 nuclei; OOD $R^2 = 0.977$ on 1,229 AME2020 nuclei absent from the original training set. Note: the extended model’s CV MAE is higher than the parsimonious figure because the Phase 2 training set includes hard exotic nuclei near the drip lines; on the original 1,319-nucleus set the extended model matches or improves on the parsimonious at every point.

The XGBoost ML Oracle (98.63% five-fold CV) is the recommended classifier for production stability classification.

Access: `tools/nuclear_be_api.py --Z [Z] --N [N] --task be|stability|both --model parsimonious|extended|both` (JSON output).

Caveat: Both analytical models cover $Z \leq 118$; predictions for unmeasured superheavy nuclei ($Z > 118$) are Category D (exploratory). Use the parsimonious model for structural interpretation and the extended model for accuracy-sensitive screening across the measured nuclear chart.

Utility note: Dedicated DFT and HFB approaches outperform both laws on the full AME dataset and should be preferred for precision work on known nuclei. The value of these laws is their analytical, dependency-free form for fast screening of unmeasured isotopes, and the GTE-derived feature motivation as a reproducible benchmark for nuclear ML researchers evaluating alternative theoretical feature sets.

3. Koide Lepton Mass Predictor (P18). A Lean 4-certified closed-form formula that predicts m_τ from m_e and m_μ alone. With PDG inputs, the prediction $m_\tau = 1776.969$ MeV is within 61 ppm (0.91σ) of the measured value; a ≤ 30 keV measurement would provide a 3.6σ test. The structural origin, $\theta = 2/9 = (N_c^2 - 1)/(4N_c^2)$ from $N_c = 3$, is machine-checked (`koide_angle_from_N_c_pure`, zero `sorry`). *Access:* formula in P18; trivial Python implementation (<10 lines). *Primary use:* precision lepton mass measurements at Belle II. *Utility note:* The empirical Koide relation has been known since 1982, but has lacked a structural derivation. P18 supplies one from first principles ($N_c = 3$ uniqueness), machine-checked in Lean 4, converting an observed numerical coincidence into a falsifiable structural prediction. For a researcher testing lepton universality or planning a Belle II m_τ measurement campaign, the formula provides a specific theoretical target ($m_\tau = 1776.969$ MeV) with a known derivation and machine-verified proof, rather than a post-hoc fit.

4. Genetic Code Sieve (P25). Evaluates any wobble-admissible codon-to-amino-acid mapping against eight independent biological viability criteria (including the generative-to-dissipative IPT criterion). No existing public tool implements the full eight-criterion combination. The standard genetic code is the unique survivor of the sieve ($z = +3.22\sigma$; 0/99,998 random wobble-admissible codes outperform it). *Access:* `papers/25_genetic_code/code/codon_sieve.py` through `codon_sieve_phase6.py` (runtime < 10 minutes; NumPy, SciPy). *Primary use:* synthetic biology, origin-of-life research, and information-theoretic analyses of the genetic code. *Utility note:* Prior computational analyses of genetic code optimality typically evaluate one or two criteria in isolation—most commonly error minimisation or polar requirement correlation. The sieve is the first tool to apply eight criteria simultaneously, including a generative-to-dissipative information balance criterion not present in any prior published framework. Because the eight criteria are evaluated jointly, the uniqueness result (0/99,998) is strictly stronger than single-criterion results: a code that satisfies one criterion well while degrading another is correctly rejected. Researchers in synthetic biology designing artificial genetic codes, or in origin-of-life studies quantifying how constrained the standard code is, can apply the sieve to any candidate mapping and receive a criterion-by-criterion breakdown, identifying precisely which biological constraints are binding.

Acknowledgements

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